

Properties of logarithms



1

a $\log 36$

b 2

c $\log_{10} 198$

d $\log_{10} 2$

e 2

f 3

g 2

h 9

i $2 + \log xy$

j $5 \log x$

k -4

l 5

m 5

n 2

o w

p $\frac{81}{64}$

2

a 2

b $\frac{48}{5}$

c 12

d $\frac{\ln x}{\ln(y^5)}$

3

a $\log_9 u + \log_9 v$

b $4 \log x$

c $7 \log_4 x$

d $\log_5 9 - \log_5 7$

e $\frac{2}{5} \log x$

f $2 \log_b x$

g $\log 3 - \log x$

h $5(\log 3 + \log x)$

i $\log 7 - 4 \log x$

j $-7 \log 5 - 7 \log x$

k $-(\log 2 + \log x)$

l $-\log x - \log y$

m $\log p + \log q - \log r$

n $-\log(3x + 7)$

o $-8 \log(3x + 4)$

p $\frac{1}{2} \log(5x + 7)$

q $6 \log(5x + 2)$

r $5 \log(x + 6)$

s $\log_2 5 + \log_2 x$

t $\log_{10} 2 - \log_{10} x$

u $\log m + \log m$

v $\log 5 + \frac{2}{3} \log x$

w $\frac{1}{3}(\log 14 + \log x)$

x $4 \log c - \frac{1}{2} \log d$

4

a $\ln 15 - \ln 8$

b $\ln p + \ln q + \ln\left(\frac{1}{r}\right)$ or $\ln p + \ln q - \ln r$

c $\ln 19 - \ln 7$

5

a $7 \log x$

b $\log(x^5 y^3)$

c $\log\left(\frac{x^8}{y^{\frac{1}{3}}}\right)$

d $\log\left(\frac{x^8}{y}\right)$

e $-35 \log_{10} 5$

f $9 \log_{10} 2$

a 2

h 2



6

a $\ln 15$

b $\ln 6$

c $\ln \left(\frac{5}{7} \right)$

d $\ln 440$

e $\ln 2$

f $\ln \left(\frac{42}{5} \right)$

7

a $\ln \left(\frac{x^6}{y^{\frac{1}{6}}} \right)$

b $\ln(15xy)$

c $5 \ln x$

d $7 \ln x$

8

a $2 \ln y$

b $-2 \ln y$

c $\frac{1}{2} \ln y$

d $\frac{1}{3} \ln y$

e $-5 \ln y$

f $\frac{2}{9} \ln y$

g $\frac{1}{2} \ln y$

h $\frac{1}{2} \ln w$

9

a $\ln \left(\frac{1}{x^2} \right)$

b $\ln \left(x^{\frac{1}{2}} \right)$

Change of base

10

a $\frac{\log_{10} 16}{\log_{10} 4}$

b $\frac{\log_{10} 0.9}{\log_{10} 3}$

c $\frac{\log_{10} \sqrt{5}}{\log_{10} 3}$

d $\frac{\log_{10} B}{\log_{10} a}$

11 $\frac{\log_4 20}{\log_4 3}$

12

a i $\frac{\log_{10} 21}{\log_{10} 8}$

ii 1.46

b i $\frac{\log_{10} \sqrt{5}}{\log_{10} 4}$

ii 0.58

c i $\frac{\log_{10} 3^3}{\log_{10} 2}$

ii 4.75

d i $\frac{\log_{10} 105}{\log_{10} \pi}$

ii 4.07